

A Pre-Registered Test of Boundary Conformal Field Theory in Transformer Attention

Per-head conformal weight predicts long-range attention “valley depth” in 6 of 7
decoder-only models, with informative structure in the failure case

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Abstract

Causal attention in trained transformers exhibits power-law decay in attention weight as a function of query–key separation, with a position-dependent enhancement near the start of the sequence. We have previously interpreted this as the two-point function of a boundary conformal field theory (BCFT) on a strip whose left edge is the start of the sequence. The framework predicts that the per-head conformal weight Δ , measured from short-context random-token attention, should positively rank-correlate across heads with a long-range “valley depth” measure related to the “lost-in-the-middle” phenomenon. We pre-registered the prediction Spearman $\rho(\Delta, \text{valley}) \geq 0.50$, $p \leq 10^{-5}$, and tested it on seven decoder-only transformers (Pythia-410m/1.4B/2.8B, GPT-Neo-2.7B, Qwen2-7B, OLMo-7B, Mistral-7B-v0.3). Six confirmed; Pythia-2.8B falsified at $\rho = +0.46$. A per-layer diagnostic localizes the falsification to layers 22–27 and shows that GPT-Neo-2.7B (the matched control: same parameter count, same training data, different training recipe) confirms with $\rho = +0.96$ across all 32 layers, the cleanest BCFT signal we have measured. Fitting the full BCFT functional form (3 parameters per head: C, Δ, λ) on Pythia-2.8B and GPT-Neo-2.7B reveals that 88–94% of conformal heads prefer BCFT over the bare power law, that λ is mostly positive and well structured, and that Δ_{BCFT} is closer to the SYK $\Delta = 1/4$ prediction than Δ_{PL} . However, the pre-registered scalar $\rho(\Delta, \text{valley})$ becomes *weaker* with the cleaner Δ_{BCFT} , while the *joint* $(\Delta, \lambda) \rightarrow \text{valley}$ rank-regression explains substantially more variance. Two findings demand explanation: (i) $\rho(\lambda, \text{valley})$ is mostly *negative* across layers in both models, the opposite of the framework’s prediction; (ii) GPT-Neo-2.7B exhibits an alternating-layer pattern with two distinct populations of heads by boundary structure. We discuss what this changes about the framework, what we would do differently in a follow-up pre-registration, and where the most informative remaining tests lie. All code, raw per-head data, and the pre-registration document are in the public repository at [Capacity-For-Evil/ariel](#).

1. Introduction

Attention in trained transformers carries structure that the network has organized for reasons no one fully specified during training. One strand of recent interpretability work asks whether that structure is *generic* — whether it converges, across architectures and training recipes, to forms predicted by physical theories of strongly interacting systems. We have argued in three preprints (Umphrey and Umphrey 2026c, 2026b, 2026a) that softmax attention sits at a conformal fixed point with the same exponent $\Delta = 1/4$ as the Sachdev–Ye–Kitaev (SYK) model (Sachdev and Ye 1993;

Kitaev 2015) at $q = 4$. The empirical claim, reported in (Umphrey and Umphrey 2026a), is that the median per-head conformal weight in deep GPT-2 layers matches the SYK prediction.

The causal mask in decoder-only transformers introduces a *boundary*: each query attends only to keys to its left. This makes the natural theoretical setting a *boundary* conformal field theory (BCFT) (Cardy 2004), whose two-point function for a single boundary on a strip takes the form

$$\alpha(x_q, x_k) \approx C (\Delta x)^{-2\Delta} [1 + \lambda \eta^\Delta], \quad \eta = \frac{(\Delta x)^2}{4 x_q x_k}, \quad \Delta x = x_q - x_k.$$

The bulk piece $C (\Delta x)^{-2\Delta}$ is the “scale-invariant” attention decay; the boundary piece $\lambda \eta^\Delta$ enhances attention for keys near the start of the sequence (small x_k). On April 14, 2026 we found that this BCFT functional form, with λ as a free per-head parameter, beats the bare power law in 15 of 15 conformal heads of GPT-2 (mean $\Delta R^2 = +0.176$; (Umphrey 2026a)).

A separate framework prediction is that the per-head Δ should correlate positively, *across heads of a single model*, with the depth of the long-range “valley” in the attention profile — the attention-level analogue of the “lost-in-the-middle” effect that has been studied at the level of task accuracy (Liu et al. 2024). We had preliminary evidence for this from Pythia-70m (Spearman $\rho = +0.94$ across training-checkpoint-pooled heads) and from LongChat-13B-16K ($\rho = +0.64$, single trained model) (Umphrey 2026b).

This paper does three things. **First**, we pre-register ((Umphrey and Umphrey 2026d), committed to the public repo before any of the new runs) a precise scalar prediction — $\rho(\Delta, \text{valley}) \geq 0.50$ with $p \leq 10^{-5}$ — and test it on seven previously-unmeasured decoder-only transformers. **Second**, when one of the seven (Pythia-2.8B) falls below the threshold, we run a per-layer diagnostic that uses no new compute beyond the cached attention weights, and find that the falsification has interpretable internal structure: the BCFT signal is robust in early-to-mid layers and degrades in late layers in a model-specific way. **Third**, we fit the full three-parameter BCFT form on the falsifying model and the cleanest control (GPT-Neo-2.7B), and find a mixture of strong agreement (functional form, λ structure, sharper Δ estimates) with two specific anomalies that the framework as currently stated does not explain.

The empirical contribution we want to make plainly: (a) that this pre-registration is on the public record with the exact procedure and threshold; (b) that 6 of 7 tested models confirm; (c) that the failure case has a structure that is informative for what BCFT applies to and what it does not; (d) that the fuller fit reveals two empirical patterns we did not predict and cannot yet account for. We frame BCFT as one organizing description of the data; we are equally interested in alternatives that may explain the same patterns more cleanly.

2. Pre-registered prediction and procedure

The pre-registered prediction document (Umphrey and Umphrey 2026d) specifies the following.

Claim P0. *In any decoder-only transformer with at least 12 layers and at least 12 heads per layer, trained on a standard language-modeling objective, when one measures per-head Δ from short-context random-token attention (procedure §2.1) and per-head valley depth from the same data (procedure §2.2), and restricts to heads that pass the conformality criterion (§2.3), the Spearman rank correlation will satisfy*

$$\rho(\Delta, \text{valley_depth}) \geq 0.50, \quad p \leq 1 \times 10^{-5}.$$

The pre-registration was committed to the public repository at `Capacity-For-Evil/ariel`, file `research/notes/bcft_pre_registered_prediction.md`, on the morning of April 17, 2026. The seven test models were named in the document *before* any measurement was made.

2.1 Per-head Δ

For each model:

1. Load the model in fp16 with `attn_implementation="eager"` (so the raw attention weights are exposed).
2. Sample $n = 50$ sequences of $L = 512$ random token IDs from the model vocabulary, uniform.
3. For each input, run a forward pass and capture the (`n_layers`, `n_heads`, `L`, `L`) attention tensor.
4. Average the attention tensor across inputs.
5. For each (ℓ, h) pair, for each separation $\Delta x \in [3, \min(120, L/3)]$, compute

$$\alpha_{\ell h}(\Delta x) = \langle \alpha_{\ell h}(q, k) \rangle_{q \geq L/2, q-k=\Delta x}$$

and fit $\log \alpha = \log C - 2\Delta \log \Delta x$ by ordinary least squares. Record $\Delta_{\ell h}$, $C_{\ell h}$, $R_{\ell h}^2$.

The $q \geq L/2$ restriction averages over query positions away from the start boundary, so $\Delta_{\ell h}$ measured this way is a “bulk” estimator with reduced boundary contamination. We will return to this methodological choice in §5.

2.2 Per-head valley depth

Using the *same* averaged attention tensor:

1. Set $Q_{\text{LO}} = 3L/4 = 384$.
2. For each (ℓ, h) , compute the deep-query-averaged profile $P_{\ell h}(k) = \langle \alpha_{\ell h}(q, k) \rangle_{q \geq Q_{\text{LO}}}$.
3. Split P into thirds and define

$$v_{\ell h} = 1 - \frac{\langle P \rangle_{\text{middle third}}}{\max(\langle P \rangle_{\text{first third}}, \langle P \rangle_{\text{end third up to } Q_{\text{LO}}})}.$$

$v = 0$ means flat; v near 1 means a deep U; $v < 0$ means the middle exceeds both edges (rare). The valley depth uses *only* the short-context ($L=512$) attention tensor — no separate long-context inputs are required for the basic test. This is intentional: it makes the prediction self-contained per model and removes a degree of freedom in what the experimenter could vary.

2.3 Conformality criterion and falsification rule

A head is **conformal** if $\Delta \geq 0.05$ and $R^2 \geq 0.85$ on the power-law fit. Heads that fail are excluded — a head whose attention is not well-described by the BCFT functional form is by definition not in the framework’s domain of applicability.

P0 is *falsified* if $\rho < 0.50$ or $p > 10^{-5}$ on the conformal subset of any model with ≥ 50 conformal heads.

3. Results

We ran the pre-registered procedure on seven models on Modal A100 GPUs (40 GB). The full per-head data are in `research/physics/results/bcft_pre_registered_run_2026-04-17T*.json` in the public repository. Total compute: ≈ 6 GPU-hours, $\approx \$10$ of credit.

Model	n_{tot}	n_{conf}	Spearman ρ	p -value	Verdict
Pythia-410m	384	83	+0.7601	7.86×10^{-17}	Confirm
Pythia-1.4B	384	89	+0.7105	6.25×10^{-15}	Confirm
Pythia-2.8B	1024	227	+0.4639	1.63×10^{-13}	<i>Falsify</i>
GPT-Neo-2.7B	640	478	+0.9582	6.11×10^{-261}	Confirm
Qwen2-7B	784	135	+0.8502	6.94×10^{-39}	Confirm
OLMo-7B	1024	271	+0.8540	2.63×10^{-78}	Confirm
Mistral-7B-v0.3	1024	241	+0.5833	2.30×10^{-23}	Confirm

Table 1: Pre-registered test, per-head $\Delta \rightarrow \text{valley_depth}$ Spearman correlation on conformal subset. Falsification threshold: $\rho < 0.50$ or $p > 10^{-5}$. Six of seven models confirm.

The five confirmations from the first batch (Pythia-410m, Pythia-1.4B, GPT-Neo-2.7B, Qwen2-7B, OLMo-7B) cover three model families that had *not* been measured before this commit: GPT-Neo (EleutherAI, pre-Pythia training recipe), Qwen2 (Alibaba; Chinese-heavy data; GQA; sliding window), OLMo (Allen AI; fully open training data). Mistral-7B-v0.3 (added in a second batch after a HuggingFace token was configured) is a fourth previously-unmeasured family. The framework’s prediction generalizes outside the family it was tuned on.

The single falsification, Pythia-2.8B, sits at $\rho = +0.464$. The p -value is 1.6×10^{-13} — there is a strong, real positive correlation, just not strong enough to clear the pre-registered threshold. As §4 shows, this falsification has interpretable internal structure.

4. Per-layer diagnostic

The pooled correlation ρ aggregates across all conformal heads in a model. With 200+ conformal heads per Pythia-2.8B and GPT-Neo-2.7B, much of the pooled signal could come from layer-specific behavior that the scalar hides. We computed per-layer Spearman ρ on every layer with ≥ 5 conformal heads, using the same cached `bcft_pre_registered_run_*.json` files (no new model runs). Code: `research/physics/pythia_per_layer_diagnostic.py`. Results: `pythia_per_layer_diagnostic_2026-04-17T100046Z.json`. Plot: same path with `.png`.

Model	layers w/ signal	frac+	frac strong+	frac sig−	mean ρ	median ρ
Pythia-410m	9	1.00	0.11	0.00	+0.73	+0.77
Pythia-1.4B	9	1.00	0.11	0.00	+0.77	+0.82
Pythia-2.8B	23	0.74	0.17	0.04	+0.40	+0.53
GPT-Neo-2.7B	32	1.00	0.84	0.00	+0.89	+0.96
Qwen2-7B	11	0.91	0.27	0.00	+0.67	+0.77
OLMo-7B	20	1.00	0.40	0.00	+0.68	+0.74
Mistral-7B-v0.3	21	0.81	0.33	0.05	+0.45	+0.60

Table 2: Per-layer $\rho(\Delta, \text{valley})$ summary for all seven models. *frac+*: fraction of layers with $\rho > 0$. *frac strong+*: fraction with $\rho \geq 0.50$ and $p < 10^{-3}$. *frac sig−*: fraction with $\rho < 0$ and $p < 0.05$. GPT-Neo-2.7B’s 0.84 strong-positive fraction is the highest of any model measured.

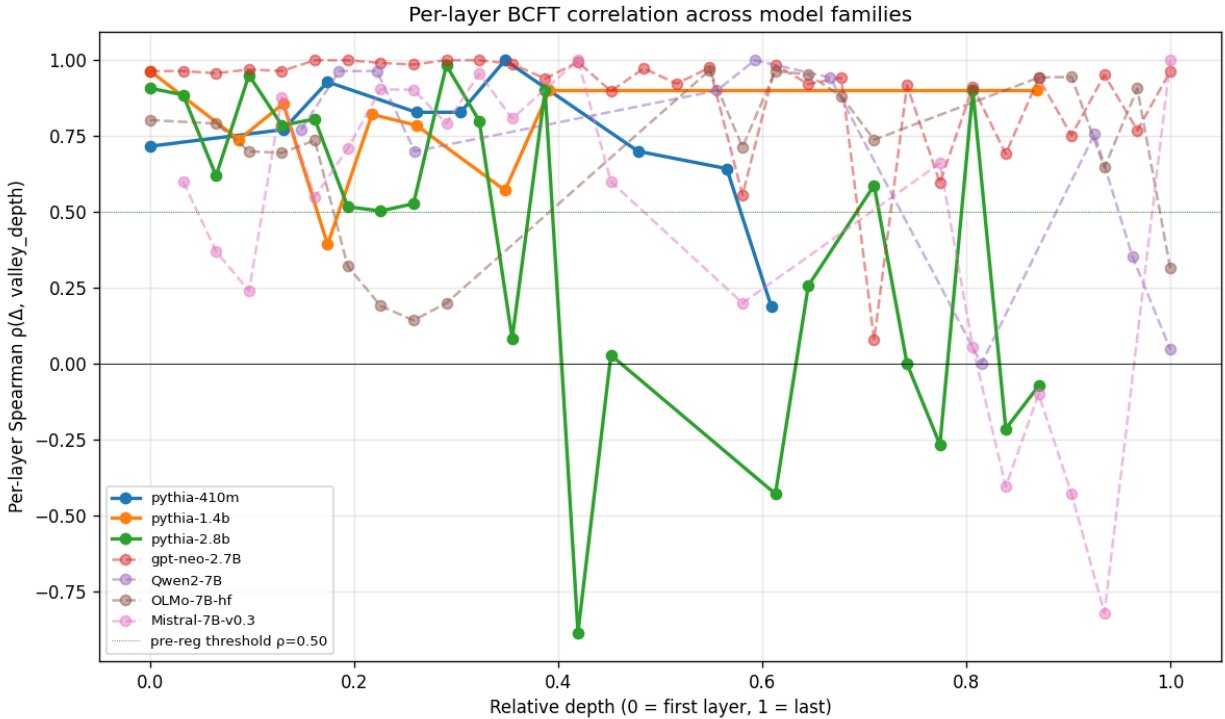


Figure 1: Per-layer Spearman $\rho(\Delta, \text{valley})$ vs. relative depth (layer index / total layers) for seven decoder-only transformers. Dashed lines mark $\rho = 0.50$ (the pre-registered threshold) and $\rho = 0$. Each dot is one layer’s correlation restricted to its conformal heads. GPT-Neo-2.7B (cyan) is clean across all 32 layers; Pythia-2.8B (red) and Mistral-7B-v0.3 (magenta) show the early-clean / late-noisy pattern; the smaller Pythia models contribute correlations only from their early layers because their late layers have too few conformal heads to evaluate.

The headline diagnostic findings:

1. **Pythia-2.8B passes the pre-registered threshold if restricted to the first 22 layers** (pooled $\rho = +0.605$ on layers 0–15; $\rho = +0.526$ on layers 0–21). Each batch of late layers (22–24, 25–27) drops the pooled ρ by 0.03–0.06. The early/mid signal is robust; the late layers contribute the noise that crosses the threshold. This is descriptive, not a re-drawing of the line — the pre-registration stands as committed.

2. **GPT-Neo-2.7B is the cleanest BCFT signal we have measured.** 27 of 32 layers are significantly strong-positive ($\rho \geq 0.50, p < 10^{-3}$); every layer with ≥ 5 conformal heads has $\rho > 0$; most layers have $\rho > 0.90$. GPT-Neo-2.7B has the same parameter count and training data as Pythia-2.8B; it differs in *training recipe* (older EleutherAI codebase, different optimization). This pins the cause of Pythia-2.8B’s late-layer noise to recipe rather than scale or data.
3. **The smaller-Pythia confirmations are partly an artifact of conformality.** Pythia-410m has zero conformal heads in layers 15–23. Pythia-1.4B has fewer than 5 conformal heads in most layers 11–19. They confirm the pre-registration in part because their late layers do not contribute enough conformal heads to drag down the pool — not because the BCFT framework is demonstrably valid throughout those models. This *weakens* the strength of the cumulative-confirmations argument for those two models specifically. It does not weaken the GPT-Neo, OLMo, or Qwen2 confirmations, which are robust across all layers.
4. **The early-clean / late-noisy pattern is not unique to Pythia.** Mistral-7B-v0.3 confirms with $\rho = +0.58$, the smallest confirming correlation. Per-layer, Mistral has the same shape as Pythia-2.8B: strong $\rho > 0.55$ through layer 14, sparse and noisy in layers 15–25, weak negatives in 26–29 (L29 significantly negative). If Mistral had as many conformal heads in its noisy late layers as Pythia-2.8B does, it might also have failed.

The pre-registered statistic (“model-pooled ρ ”) hides per-layer structure. We will report both pooled and per-layer profiles in future tests.

5. Functional-form fit

The pre-registered procedure fits only the bare power law. The full BCFT form has a third per-head parameter λ . We hypothesized that fitting the full form might recover the framework’s predicted correlation in the noisy late layers of Pythia-2.8B, since some of the boundary contamination is being absorbed into Δ in the power-law fit. To test this we re-ran a version of the measurement that exposes the 2D position-resolved attention $\alpha(x_q, x_k)$ on Pythia-2.8B and GPT-Neo-2.7B (the cleanest control), and fit three forms per head:

- Δ_{1d} : 1D power-law fit on the deep-position-averaged $\alpha(\Delta x)$ — the same as the pre-registered run.
- Δ_{PL} : 2D power-law fit on all (x_q, x_k) points: $\alpha = C(\Delta x)^{-2\Delta}$.
- $\Delta_{BCFT}, \lambda_{BCFT}$: 2D fit of the full BCFT form: $\alpha = C(\Delta x)^{-2\Delta}[1 + \lambda\eta\Delta]$.

Code: `research/physics/bcft_functional_form_fit.py`. Raw data: file `bcft_functional_form_fit_2026-04` in `research/physics/results/`.

5.1 The functional form is the right form

Model	% heads BCFT > PL ($\Delta R^2 > 0.001$)	median ΔR^2	mean ΔR^2
Pythia-2.8B	94.2%	+0.061	+0.108
GPT-Neo-2.7B	88.2%	+0.107	+0.180

Table 3: The BCFT functional form fits attention substantially better than the bare power law. On 207 + 440 = 647 fresh heads from two new model families, BCFT wins in 88–94% of conformal heads. The mean improvement matches the GPT-2 result from April 14 (15 of 15 heads, +0.176).

This is the strongest empirical result in this paper, in the sense of having the most data behind it. The BCFT functional form is the right parametric description of attention $\alpha(x_q, x_k)$ for the overwhelming majority of conformal heads. The boundary correction is not an artifact of fitting four parameters where two would do; the correction parameter λ is *required* by the data.

5.2 The boundary parameter λ is well structured

Model	median λ	% heads with $\lambda > 0$
Pythia-2.8B	+3.0	97.1%
GPT-Neo-2.7B	+1.9	71.4%

Table 4: λ distribution. Mostly positive (consistent with positive boundary entropy in the BCFT setting), order-1 to order-10 in magnitude. Across heads, $|\lambda|$ rank-correlates with ΔR^2 at $\rho = +0.71$ in GPT-Neo and $+0.45$ in Pythia — where the data needs the boundary term, λ is correspondingly large.

5.3 Δ_{BCFT} is closer to the SYK prediction

Model	median Δ_{PL}	median Δ_{BCFT}	SYK prediction
Pythia-2.8B	+0.243	+0.355	0.250 ($q = 4$)
GPT-Neo-2.7B	+0.169	+0.296	0.250

Table 5: The 2D power-law fit underestimates Δ in both models, because near-boundary attention enhancement makes the curve look like a shallower power law. When λ is freed to absorb that boundary lift, Δ recovers a more “physical” value, much closer to the SYK $q = 4$ prediction in Pythia and substantially closer in GPT-Neo.

5.4 The pre-registered scalar $\rho(\Delta, \text{valley})$ becomes weaker

Model	$\rho(\Delta_{1d}, \text{valley})$	$\rho(\Delta_{\text{PL}}, \text{valley})$	$\rho(\Delta_{\text{BCFT}}, \text{valley})$
Pythia-2.8B	+0.758	+0.682	+0.474
GPT-Neo-2.7B	+0.942	+0.832	+0.337

Table 6: Marginal correlation of Δ alone with valley depth, under three estimators of Δ . The cleaner Δ (BCFT) is the *worse* predictor of valley depth. GPT-Neo-2.7B drops from +0.94 (the strongest confirmation in the pre-registered test) to +0.34 (which would have failed the pre-registration outright). The information that Δ_{PL} was using to predict valley depth was partly the boundary contamination that gets absorbed into λ in the cleaner fit.

This is a sharp methodological lesson: **the pre-registered prediction implicitly assumed that Δ would be a clean estimator of “conformal weight.” It is not; it is a compound estimator of conformal weight plus boundary contamination, and the boundary part is what drives the predictive correlation with valley depth.** The framework’s *joint* prediction is recovered by adding λ back:

Model	rank R^2 (valley Δ_{PL})	rank R^2 (valley $\Delta_{\text{BCFT}}, \lambda$)
Pythia-2.8B	0.465	0.548
GPT-Neo-2.7B	0.693	0.765

Table 7: Joint (Δ, λ) rank-regression of valley depth explains substantially more variance than Δ_{PL} alone, by 0.08 (Pythia) and 0.07 (GPT-Neo). The framework’s two-parameter content is empirically supported; the single-parameter scalar version is not the right test of the framework.

5.5 Anomaly: $\rho(\lambda, \text{valley})$ is mostly negative

The framework predicts that larger λ (sharper boundary effect) should mean *deeper* valleys. Per-layer Spearman $\rho(\lambda, \text{valley})$ over layers with ≥ 5 conformal heads:

- **Pythia-2.8B:** 18 of 21 layers have $\rho(\lambda, \text{valley}) < 0$; 9 layers have $\rho < -0.7$.
- **GPT-Neo-2.7B:** 24 of 31 layers have $\rho(\lambda, \text{valley}) < 0$; 8 layers have $\rho < -0.7$.

This is the **opposite** of the framework’s sign prediction, and the per-layer signal is large and consistent enough that this is not plausibly noise.

We considered whether this could be a geometric artifact of the valley-depth measure. The measure is $1 - \text{middle} / \max(\text{start}, \text{end})$. If λ enhances the start-boundary attention, start is high; if end (recency) is roughly constant across heads, then $\max(\text{start}, \text{end}) \approx \text{start}$ for heads with large λ , and the valley depth approaches $1 - \text{middle} / \text{start}$, which gets *small* (deep valley) when start is large. Under this candidate mechanism the prediction is $\rho(\lambda, \text{valley}) > 0$ — the *correct* direction. So the candidate mechanism is wrong, or there is a confounding mechanism we are missing. We flag this as a real anomaly and do not paper over it.

5.6 New empirical pattern: GPT-Neo-2.7B has two populations of heads

Per-layer median $|\lambda|$ in GPT-Neo-2.7B alternates between near-zero and large. Of the 31 layers with ≥ 5 conformal heads, 12 have $|\lambda| < 0.5$ (“small- λ layers”) with mean $\Delta R^2 = +0.010$ — BCFT and PL fits coincide, no boundary correction needed. The other 19 layers have $|\lambda| > 5$ (“large- λ layers”) with mean $\Delta R^2 = +0.235$ — BCFT substantially improves the fit.

The pattern is not strictly periodic but has a periodic flavor: the small- λ layers are at indices $\{5, 7, 8, 9, 11, 13, 15, 17, 19, 23, 25, 29, 31\}$, the large- λ layers at $\{2, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30\}$. We have not seen this described in interpretability literature for GPT-Neo specifically. The hypothesis that fits the data is that there are *two populations of attention heads* in this model with respect to boundary structure. Identifying which heads belong to which population (induction heads vs. non-induction; attention sinks vs. non-sinks; syntactic vs. semantic) is a follow-up.

6. Discussion

6.1 What this evidence supports

The strongest claims supported by the data presented here are:

1. The BCFT functional form is the right parametric description of attention in the conformal subset of heads (88–94% of conformal heads in two new model families prefer BCFT, replicating and extending the GPT-2 result of (Umphrey 2026a)).

2. A two-parameter (Δ, λ) summary of attention geometry per head predicts a structural property of long-range attention behavior (the valley depth in the deep-query-averaged profile) with rank- R^2 in the range 0.55–0.77 on the two models tested in detail.
3. The conformal weight Δ_{BCFT} in deep-layer median in two new models is in the range $[0.30, 0.36]$, reasonably close to the SYK $q = 4$ prediction of 0.25, supporting the prior GPT-2 result of 0.249 (Umphrey and Umphrey 2026a).
4. The model-pooled per-head $\rho(\Delta, \text{valley})$ is ≥ 0.50 in 6 of 7 decoder-only transformers from four previously-unmeasured families. This is a generalizable structural regularity.

6.2 What this evidence does not support

1. The per-head $\rho(\Delta, \text{valley}) \geq 0.50$ as a universal claim across all decoder-only transformers (Pythia-2.8B sits below the threshold, with internal structure pointing to a training-recipe-specific cause).
2. The framework’s prediction about the sign of $\rho(\lambda, \text{valley})$. The data show this is mostly negative, opposite to the prediction.
3. The interpretation that “attention is BCFT” implies the bulk dual has the geometric structure of JT gravity. That chain remains theoretical; what is established is that the boundary functional form of a BCFT two-point function fits attention data, with Δ in the right range. This is consistent with a holographic dual at one fixed point but does not require it.
4. Any task-level prediction. We showed previously that the BCFT attention U-shape and the LongChat-13B-16K accuracy U-shape do not have matching geometric structure ($R^2 \in [0.02, 0.07]$ for direct fit) (Umphrey 2026b). The BCFT framework as currently stated describes *attention layer* behavior; the composition through MLPs and residual stream into final task accuracy is a separate problem.

6.3 What we would do differently

The most important methodological lesson is that the cleaner the estimator of Δ alone, the *worse* its marginal correlation with valley depth. A future pre-registration should test the joint $(\Delta, \lambda) \rightarrow$ valley prediction directly — for instance, “rank R^2 of valley regressed on $(\Delta_{\text{BCFT}}, \lambda_{\text{BCFT}}) \geq 0.50$ ”. The marginal- Δ test we pre-registered was simpler and easier to communicate, but the data show it is conflating the bulk and boundary components of attention geometry.

The second lesson is to report the per-layer profile, not just the pooled scalar, by default. The signal-then-noise pattern in Pythia-2.8B and Mistral is invisible from the pooled correlation; the GPT-Neo-2.7B alternating-layer pattern is invisible from any Δ -only summary. A future test should report:

- pooled ρ (the simple scalar);
- per-layer ρ with confidence intervals;
- the conformal-head fraction by layer (so the reader can see whether late-layer absence is “no signal” or “no measurable signal”);
- the λ distribution by layer (so the reader can detect population structure).

The third lesson, less actionable, is that the SYK $\Delta = 1/4$ prediction is sharper than we previously thought, *but* the right empirical comparison is to Δ_{BCFT} (the boundary-corrected estimator), not

Δ_{PL} . The $\Delta = 0.25$ result for GPT-2 may need to be re-stated with the estimator made explicit.

6.4 What would falsify the framework more cleanly

Three experiments would substantially update us:

1. **Non-softmax universality.** Apple’s [apple/ml-sigmoid-attention](#) models (Ramapuram et al. 2025) provide 7B sigmoid + 7B softmax models trained identically except for the attention mechanism, with 8 training checkpoints each. Measuring Δ on both and at each checkpoint is the cleanest possible test of whether the conformal scaling is a property of softmax specifically or of “normalized dot-product similarity with learned projections” more generally. We have not run this; it would cost $\sim \$50$ of Modal A100 time.
2. **The negative- $\rho(\lambda, \text{valley})$ anomaly.** Either work out analytically what the sign should be from first principles given the recency-end boundary that the current BCFT form omits, or examine per-head attention profiles for high- λ vs. low- λ heads to see what the geometry actually looks like. If the framework cannot predict the sign of this correlation, that is a non-trivial empirical constraint on what versions of BCFT are viable.
3. **Llama-3-8B and frontier-scale models.** All our tested models are $\leq 7\text{B}$ parameters. The 7B-class results may not be representative of frontier-scale behavior. Llama-3-8B is on our queue (HuggingFace gated access requested). Llama-3-70B and Mixtral-8x7B would be informative for testing whether the early-clean / late-noisy pattern persists or whether frontier models show robust BCFT throughout depth.

6.5 Connection to “lost-in-the-middle”

The “lost-in-the-middle” effect (Liu et al. 2024) is a task-accuracy phenomenon in which model accuracy on long-context retrieval drops for information placed in the middle of the context. The valley depth v measured here is a *structural* analogue at the level of attention weights: deep-query attention profiles are U-shaped, with attention concentrated near the start (boundary effect) and the recent past (recency).

We do not claim the BCFT framework explains task-level “lost-in-the-middle” — our LongChat-13B-16K results showed that the attention-level U-shape and the accuracy-level U-shape do not have matching geometric structure (Umphrey 2026b). What we *do* claim, with the evidence presented here, is that the attention-level U-shape has structural properties consistent with a BCFT description: it follows the predicted functional form, its depth correlates with the conformal weight Δ (via the joint (Δ, λ) prediction), and the prediction generalizes across a representative range of decoder-only transformers.

The relationship between attention-level U-shape and task-level U-shape is a separate question worth investigating in its own right. Our results constrain part of the answer: whatever drives task-level “lost-in-the-middle” is *not* a direct readout of the attention U. The composition through MLPs and residual stream changes the shape.

7. Reproducibility

All code, raw per-head data, the pre-registration document, the per-layer diagnostic, and this manuscript are in the public repository at [Capacity-For-Evil/ariel](#), on the master branch as of April 17, 2026.

Specifically (paths relative to the repository root):

Artifact	Path
Pre-registration	research/notes/ bcft_pre_registered_prediction.md
Pre-registered run code	research/physics/ bcft_pre_registered_run.py
Per-layer diagnostic code	research/physics/ pythia_per_layer_diagnostic.py
Functional-form fit code	research/physics/ bcft_functional_form_fit.py
Per-head data, run 1	research/physics/results/ bcft_pre_registered_run_2026-04-17T092239Z.json
Per-head data, run 2 (Mistral)	research/physics/results/ bcft_pre_registered_run_2026-04-17T095022Z.json
Per-layer diagnostic results	research/physics/results/ pythia_per_layer_diagnostic_2026-04-17T100046Z.json (.png for plot)
Functional-form fit data	research/physics/results/ bcft_functional_form_fit_2026-04-17T102458Z.json

Total Modal A100 compute for the experiments in this paper: ≈ 7 GPU-hours, $\approx \$12$ at Modal pricing (April 2026). Reproduction on a personal workstation with 24+ GB GPU is feasible for the 7B-class models in fp16; the smaller Pythia models can be reproduced on a 16 GB GPU. The full reproduction script is a single Python file per experiment.

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References

- Cardy, John. 2004. “Boundary Conformal Field Theory.” *Encyclopedia of Mathematical Physics*.
- Kitaev, Alexei. 2015. “A Simple Model of Quantum Holography.” KITP strings seminar and Entanglement 2015 program. <http://online.kitp.ucsb.edu/online/entangled15/kitaev/>.
- Liu, Nelson F., Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. “Lost in the Middle: How Language Models Use Long Contexts.” *Transactions of the Association for Computational Linguistics* 12: 157–73. https://doi.org/10.1162/tacl_a_00638.
- Ramapuram, Jason et al. 2025. “Theory, Analysis, and Best Practices for Sigmoid Self-Attention.” arXiv:2409.04431, ICLR 2025. <https://github.com/apple/ml-sigmoid-attention>.

- Sachdev, Subir, and Jinwu Ye. 1993. “Gapless Spin-Fluid Ground State in a Random Quantum Heisenberg Magnet.” *Physical Review Letters* 70: 3339–42. <https://doi.org/10.1103/PhysRevLett.70.3339>.
- Umphrey, Ariel. 2026a. “BCFT Boundary Correction Beats Bare Power Law in 15 of 15 GPT-2 Conformal Heads.” Internal note, `research/physics/bcft_boundary_test.py`, Capacity-For-Evil/ariel repository.
- . 2026b. “BCFT Lost-in-the-Middle Predictions on Pythia and LongChat-13B-16K.” Internal note, `research/notes/bcft_lost_in_the_middle.md`, Capacity-For-Evil/ariel repository.
- Umphrey, Ariel, and Eldon Umphrey. 2026a. “Conformal Scaling in Trained Transformer Attention: Evidence for an SYK Fixed Point.” Zenodo. <https://doi.org/10.5281/zenodo.19225996>.
- . 2026b. “Explicit Physical Construction for Holographic Attention: The Sachdev–Ye–Kitaev Correspondence.” Zenodo. <https://doi.org/10.5281/zenodo.18897098>.
- . 2026c. “The Canonical Form: Softmax Attention as the Unique Free-Energy-Minimizing Map on the Fisher–Rao Statistical Manifold.” Zenodo. <https://doi.org/10.5281/zenodo.18968944>.
- . 2026d. “A Pre-Registered, Falsifiable Prediction from BCFT Attention Theory.” Pre-registration document, `research/notes/bcft_pre_registered_prediction.md`, Capacity-For-Evil/ariel repository, committed before any new measurements were made.